# CAPSTONE PROJECT REVIEW PROGRESS RECORD

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| **Review:** | **Review–3** | **Date of Review:** | **6 November 2026** |
| **Project Title:** | **Sensitivity-Guided Layer-Wise Approximate Multiplier Mapping for FPGA-Based CNN Accelerators** | | |
| **Cohort:** | _____ | **Project Guide:** | Dr. Arun Nimmala, Associate Professor, Dept. of ECE |

**Student Details:**

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## Brief Description of the Project

*Briefly explain the premise of the project, the problem being addressed and the proposed approach. (less than 200 words)*

Convolutional Neural Network (CNN) edge accelerators are constrained by the high area and power dissipation of Multiply-Accumulate (MAC) datapaths, where 8x8 exact multipliers dominate silicon cost (68 LUTs on Xilinx Zynq-7020). While approximate computing trades controlled arithmetic error for hardware savings, uniform approximation causes severe accuracy degradation or negligible gains. Existing layer-wise mapping methods rely on computationally prohibitive evolutionary search (ALWANN: 7.5 CPU-days) or GPU retraining loops (TRAM). This project proposes a lightweight, single-pass sensitivity metric computed in 0.2 seconds combining static weight variance and dynamic loss gradients via a 256-sample calibration probe without retraining. Layers are ranked such that error-vulnerable early layers receive exact multipliers while noise-resilient deep layers receive aggressive approximate multipliers. The heterogeneous design is mapped onto synthesizable Verilog Processing Elements (PEs) and deployed on a PYNQ-Z2 FPGA board via AXI4-Stream DMA, achieving up to 32.5% multiplier area reduction while preserving baseline classification accuracy.

## Progress Achieved up to This Review

*Briefly mention the work completed, implementation/simulation/experimental progress, results obtained and other major milestones achieved up to this review.*

- **NSGA-II Multi-Objective Optimization Benchmark:** Implemented evolutionary search across all 256 configurations optimizing Accuracy Loss vs. LUT Cost. Confirmed our sensitivity-guided assignment lands at the Pareto knee in 0.2s, achieving a >3,000,000x speedup over genetic search (ALWANN: 7.5 CPU-days).
- **Complete Synthesizable Accelerator RTL:** Designed full Verilog datapath (axm_8x8.v, mac_unit.v, conv_pe.v, layer_controller.v, cnn_accelerator_top.v) with 2-bit runtime mode control and AXI4-Stream interfaces. Verified across 262,144 exhaustive test vectors with 100% pass (0 mismatches).
- **Automated Vivado Block Design:** Developed vivado_block_design.tcl automating Zynq-7020 PS/PL integration, AXI SmartConnect, and AXI DMA engine (MM2S & S2MM channels) clocked at 100 MHz.
- **PYNQ-Z2 On-Board Deployment Pipeline:** Extracted INT8 quantized weights (weights_int8.npy), built Python on-board driver (pynq_inference.py), and authored an interactive 7-cell Jupyter notebook (pynq_cnn_demo.ipynb) for hardware telemetry.
- **Publication Readiness & Code Release:** Generated all 7 IEEE publication figures, drafted 6-page IEEE manuscript, compiled comprehensive 20-slide presentation deck with embedded speaker notes, and published repository to GitHub.

**Signature of Project Guide:** _____ **Date:** _____